

Willie Xu

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SKILLS

Cloud & Infrastructure: AWS (EC2, VPC, IAM), Linux Servers, Active Directory

DevSecOps & IaC: Terraform, Git, GitHub Actions, CI/CD Pipelines, Docker

Security & Compliance: SAST (tfsec), Vulnerability Assessment, Network Security, Firewall Configuration (iptables), Log Analysis

Programming & Scripting: Python, Bash/Shell Scripting

PROJECTS/EXPERIENCE

Foundations Lab Final

Feb 2026 – Feb 2026

- Executed Multi-Phase Security Assessment: Performed active network reconnaissance and active scanning across multiple subnets (172.100.0.0/24, 172.80.0.0/24, 172.60.0.0/24) to identify active hosts, exposed services, and security misconfigurations.
- Rapid Triage & System Hardening: Remediated three compromised servers by terminating unauthorized background processes, restricting dangerous file permissions, and configuring strict firewall rules to eliminate network exposures.
- Credential Brute-Forcing & Log Forensics: Simulated an SSH brute-force attack using Hydra and a customized wordlist, successfully cracked credentials, and parsed system authentication logs to extract attacker IPs and exact timestamps for incident tracking.
- Web Exploit Analysis & Perimeter Defense: Exploited a command injection vulnerability in a web application using cURL and Netcat to establish a reverse shell, subsequently analyzed access logs for malicious User-Agents, and engineered iptables drop rules to lockdown compromised containers.

Foundations Lab Final

Aug 2026 – Aug 2026

- Infrastructure as Code (IaC) Architecture: Designed and deployed a secure, automated AWS cloud stack using Terraform, provisioning custom VPCs (10.0.0.0/16), public subnets, internet gateways, and route tables.
- Cloud Security & Network Hardening: Configured granular security groups enforcing least-privilege ingress rules (limiting SSH access to specific trusted IPs while permitting HTTP traffic) and automated EC2 Apache web server provisioning via user_data scripts.
- DevSecOps Quality Gate & SAST Integration: Engineered a custom GitHub Actions CI/CD workflow (security-scan.yml) integrating tfsec static application security testing (SAST) to automatically scan Infrastructure-as-Code definitions for misconfigurations on every git push.
- Policy Enforcement & Compliance: Configured hard security gates with soft-fail: false to break builds upon detecting insecure security groups or missing encryption standards prior to deployment.

EDUCATION

THE KNOWLEDGE HOUSE, Bronx, NY

Jan 2026 – Present

Innovation Fellowship - [Cyber Security]

- Enhance skills in cybersecurity in a competitive, project-based fellowship with 400+ hours of technology and professional development instruction
- Developed core AI literacy to understand market practices and how to use AI as a learning and productivity tool.
- Deliver virtual presentations of final projects to tech professionals, funders, and community stakeholders

Stony Brook University, Stony Brook, NY

Aug 2016 – Aug 2020

B.S., Computer Engineering

- Relevant courses: Operating Systems, Data Structures, Algorithms